

Research Notes on Prompt engineering

Prompt engineering

>> Section 1

Using gradient descent to search for prompts In "prefix-tuning", "prompt tuning", or "soft prompting", floating-point vectors are searched directly by gradient descent to maximize the log-likelihood on outputs. An earlier result uses the same idea of gradient descent search, but is designed for masked language models like BERT, and searches only over token sequences, rather than numerical vectors. Formally, it searches for $\arg \max_{X \sim \sum_i \log \Pr[Y_i | X \sim * X_i]}$ where $X \sim$ ranges over token sequences of a specified length.

>> Section 2

A text-to-image prompt commonly includes a description of the subject of the art, the desired medium (such as digital painting or photography), style (such as hyperrealistic or pop-art), lighting (such as rim lighting or crepuscular rays), color, and texture. Word order also affects the output of a text-to-image prompt. Words closer to the start of a prompt may be emphasized more heavily.

>> Section 3

Prompt engineering is the process of structuring natural language inputs (known as prompts) to produce specified outputs from a generative artificial intelligence (GenAI) model. Context engineering is the related area of software engineering that focuses on the management of non-prompt contexts supplied to the GenAI model, such as metadata, API tools, and tokens. It can also be defined as the practice of designing and refining input instructions given to a generative AI model to produce more accurate, relevant, or useful outputs. Effective prompt engineering involves understanding how a model interprets language, and may include techniques such as few-shot prompting, chain-of-thought prompting, and role assignment. It is increasingly considered a skill for working with large language models (LLMs) in both research and professional contexts. During the 2020s AI boom, prompt engineering became regarded as a business capability across corporations and industries. Employees with the title prompt engineer were hired to create prompts that would increase productivity and efficacy, although the individual title has since lost traction amid AI models that produce better prompts than humans and corporate training in prompting for general employees. Common prompting techniques include multi-shot, chain-of-thought, and tree-of-thought prompting, as well as the use of assigning roles to the model. Automated prompt generation methods, such as retrieval-augmented generation (RAG), provide for greater accuracy and a wider scope of functions for prompt

engineers. Prompt injection is a type of cybersecurity attack that targets machine learning models through malicious prompts.

>> Section 4

Chain-of-thought (CoT) prompting is a technique that allows large language models (LLMs) to solve a problem as a series of intermediate steps before giving a final answer. In 2022, Google Brain reported that chain-of-thought prompting improves reasoning ability by inducing the model to answer a multi-step problem with steps of reasoning that mimic a train of thought. Chain-of-thought techniques were developed to help LLMs handle multi-step reasoning tasks, such as arithmetic or commonsense reasoning questions. When applied to PaLM, a 540 billion parameter language model, according to Google, CoT prompting significantly aided the model, allowing it to perform comparably with task-specific fine-tuned models on several tasks, achieving state-of-the-art results at the time on the GSM8K mathematical reasoning benchmark. It is possible to fine-tune models on CoT reasoning datasets to enhance this capability further and stimulate better interpretability. As originally proposed by Google, each CoT prompt is accompanied by a set of input/output examples—called exemplars—to demonstrate the desired model output, making it a few-shot prompting technique. However, according to a later paper from researchers at Google and the University of Tokyo, simply appending the words "Let's think step-by-step" was also effective, which allowed for CoT to be employed as a zero-shot technique.

>> Section 5

Retrieval-augmented generation (RAG) Retrieval-augmented generation is a technique that enables GenAI models to retrieve and incorporate new information. It modifies interactions with an LLM so that the model responds to user queries with reference to a specified set of documents, using this information to supplement information from its pre-existing training data. This allows LLMs to use domain-specific and/or updated information. RAG improves large language models by incorporating information retrieval before generating responses. Unlike traditional LLMs that rely on static training data, RAG pulls relevant text from databases, uploaded documents, or web sources. By dynamically retrieving information, RAG enables AI to generate more accurate responses and fewer AI hallucinations without frequent retraining.

>> Section 6

Prompt A prompt is some natural language text that describes and prescribes the task that an artificial

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intelligence (AI) should perform. A prompt for a text-to-text language model can be a query, a command, or a longer statement referencing context, instructions, and conversation history. The process of prompt engineering may involve designing clear queries, refining wording, providing relevant context, specifying the style of output, and assigning a character for the AI to mimic in order to guide the model toward more accurate, useful, and consistent responses. When communicating with a text-to-image or a text-to-audio model, a typical prompt contains a description of a desired output such as "a high-quality photo of an astronaut riding a horse" or "Lo-fi slow BPM electro chill with organic samples". Prompt engineering may be applied to text-to-image models to achieve a desired subject, style, layout, lighting, and aesthetic.

>> Section 7

Multi-shot A prompt may include a few examples for a model to learn from in context, an approach called few-shot learning. For example, the prompt may ask the model to complete "maison → house, chat → cat, chien →", with the expected response being dog.

>> Section 8

In 2022, text-to-image models like DALL-E 2, Stable Diffusion, and Midjourney were released to the public. These models take text prompts as input and use them to generate images. Early text-to-image models typically do not understand negation, grammar and sentence structure in the same way as large language models, and may thus require a different set of prompting techniques. The prompt "a party with no cake" may produce an image including a cake.

>> Section 9

Textual inversion and embeddings For text-to-image models, textual inversion performs an optimization process to create a new word embedding based on a set of example images. This embedding vector acts as a "pseudo-word" which can be included in a prompt to express the content or style of the examples.